1112 - b)

$$\int_0^3 |x^2-3x+2|\,dx = \int_0^1 (x^2-3x+2)\,dx + \int_1^2 (-x^2+3x-2)\,dx + \int_2^3 (x^2-3x+2)\,dx =$$

$x^2-3x+2=0$

$(x-1)\cdot(x-2)=0$

		1		2	
$x-1$	$-$		$+$		$+$
$x-2$	$-$		$-$		$+$
	$+$		$-$		$+$

$$= \left[\frac{x^3}{3}-\frac{3x^2}{2}+2x\right]_0^1 + \left[-\frac{x^3}{3}+\frac{3x^2}{2}-2x\right]_1^2 + \left[\frac{x^3}{3}-\frac{3x^2}{2}+2x\right]_2^3 =$$

$$= \left(\left(\frac{1}{3}-\frac{3}{2}+2\right)-0\right) + \left(\left(\frac{-8}{3}+\frac{12}{2}-4\right)-\left(\frac{-1}{3}+\frac{3}{2}-2\right)\right) + \left(\left(\frac{27}{3}-\frac{27}{2}+6\right)-\left(\frac{8}{3}-\frac{12}{2}+4\right)\right) =$$

$$= \left(\frac{1}{3}-\frac{3}{2}+2\right) + \left(\frac{-7}{3}+\frac{9}{2}-2\right) + \left(\frac{19}{3}-\frac{15}{2}+2\right) = \frac{5}{6}+\frac{1}{6}+\frac{5}{6} = \underline{\underline{\frac{11}{6}}}$$

11/3-a)

$$\int_0^{\frac{\pi}{6}} (x+2)\sin \overset{3x}{\cancel{\tfrac{3}{x}}}\, dx = \left| \begin{matrix} u = x+2 & v' = \sin 3x \\ u' = 1 & v = -\frac{1}{3}\cos \overset{3x}{\cancel{\tfrac{3}{x}}} \end{matrix} \right| \overset{PP}{=} \left[(x+2)\cdot\left(-\frac{1}{3}\cos \overset{3x}{\cancel{\tfrac{3}{x}}}\right)\right]_0^{\frac{\pi}{6}} + \int_0^{\frac{\pi}{6}} \frac{1}{3}\cos 3x\, dx =$$

$$= \left[(x+2)\cdot\left(-\frac{1}{3}\cos 3x\right)\right]_0^{\frac{\pi}{6}} + \left[\frac{1}{9}\sin 3x\right]_0^{\frac{\pi}{6}} = \left(\left(\left(\frac{\pi}{6}+2\right)\cdot\left(-\frac{1}{3}\cos 3\frac{\pi}{6}\right)\right) - \left(2\cdot -\frac{1}{3}\cos 0\right)\right) +$$

$$+ \left(\frac{1}{9}\sin 3\frac{\pi}{6} - \frac{1}{9}\sin 0\right) = \left(0 + \frac{2}{3}\right) + \left(\frac{1}{9} - 0\right) = \frac{2}{3} + \frac{1}{9} = \underline{\underline{\frac{7}{9}}}$$

11/5-c)

$$\int_{e^{-1}}^{e} \frac{\ln x + 1}{x(\ln^2 x + 1)}\, dx = \left| \begin{matrix} u = \ln x \\ du = \frac{1}{x}\, dx \\ e \rightsquigarrow 1 \\ e \rightsquigarrow -1 \end{matrix} \right| = \int_{-1}^{1} \frac{u+1}{u^2+1}\, du = \int_{-1}^{1} \left(\frac{u}{u^2+1} + \frac{1}{u^2+1}\right) du =$$

$$= \int_{-1}^{1} \left(\frac{1}{2}\frac{2u}{u^2+1} + \frac{1}{u^2+1}\right) du = \left[\frac{1}{2}\ln|u^2+1| + \operatorname{arctg} u\right]_{-1}^{1} = \left(\frac{1}{2}\ln 2 + \operatorname{arctg} 1\right) - \left(\frac{1}{2}\ln 2 + \operatorname{arctg}(-1)\right) =$$

$$= \operatorname{arctg} 1 - \operatorname{arctg}(-1) = \frac{\pi}{4} + \frac{\pi}{4} = \underline{\underline{\frac{\pi}{2}}}$$

11/6-a)

$$\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \sin^4 2x \, dx = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \left(\frac{1-\cos 4x}{2}\right)^2 = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{4}(1-\cos 4x)^2 = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{4}\left(\cos^2 4x - 2\cos 4x + 1\right) dx =$$

$$= \frac{1}{4}\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} (\cos^2 4x - 2\cos 4x + 1)\,dx = \frac{1}{4}\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \left(\frac{1}{2}\cos 8x + \frac{1}{2}\right) dx - \frac{1}{2}\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \cos 4x \, dx + \frac{1}{4}\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} 1\,dx =$$

$$= \frac{1}{8}\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} (\cos 8x + 1)\,dx - \frac{1}{2}\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \cos 4x \, dx + \frac{1}{4}\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} 1\,dx = \frac{1}{8}\left[x + \frac{1}{8}\sin(8x)\right]_{\frac{\pi}{8}}^{\frac{3\pi}{8}} - \frac{1}{2}\left[\frac{1}{4}\sin 4x\right]_{\frac{\pi}{8}}^{\frac{3\pi}{8}} +$$

$$+ \frac{1}{4}\left[x\right]_{\frac{\pi}{8}}^{\frac{3\pi}{8}} = \frac{1}{8}\left(\left(\frac{3\pi}{8} + \frac{1}{8}\sin(3\pi)\right) - \left(\frac{\pi}{8} + \frac{1}{8}\sin \pi\right)\right) - \frac{1}{2}\left(\frac{1}{4}\sin\frac{3\pi}{2} - \frac{1}{4}\sin\frac{\pi}{2}\right) + \frac{1}{4}\left(\frac{3\pi}{8} - \frac{\pi}{8}\right) =$$

$$= \frac{1}{8}\cdot\frac{\pi}{4} - \frac{1}{2}\cdot\frac{-1}{2} + \frac{1}{4}\cdot\frac{\pi}{4} = \frac{\pi}{32} + \frac{1}{4} + \frac{\pi}{16} = \frac{3\pi}{32} + \frac{1}{4}$$

12/3-b $\quad x-1 \neq 0 \Rightarrow x \neq 1$

$$\int_0^2 \frac{2}{\sqrt[3]{x-1}}\,dx = \int_0^1 \frac{2}{\sqrt[3]{x-1}} + \int_1^2 \frac{2}{\sqrt[3]{x-1}} = \lim_{x\to 1^-}\left(\left[3(x-1)^{2/3}\right]_0^x\right) + \lim_{x\to 1^+}\left(\left[3(x-1)^{2/3}\right]_x^2\right) =$$

$$= \cancel{\lim_{x\to 1^-}\left(3^{2/3} - (3x+1)^{2/3}\right) + \lim} \cancel{\lim_{x\to 1^-}\left(3(x+1)^{2/3} - 3^{2/3}\right) + \lim_{x\to 1^+}\left(9^{2/3} - 3(x+1)^{2/3}\right)} =$$

$$= \cancel{\left(6^{2/3} - 3^{2/3}\right)} + \left(9^{2/3} - 6^{2/3}\right) = 9^{2/3} - 3^{2/3}$$

$$= \cancel{3[x-1]^{2/3} + 3[\ldots]} \; 3\cdot\left((1-1)^{2/3} + 1\right) + 3\cdot\left((1-1)^{2/3} - 1\right) = 0$$

7215 b

$$\int_{\ln 4}^{\infty} \frac{4e^x}{e^{2x}-4}\,dx = \left| \begin{array}{l} t = e^x \\ dt = e^x dx \\ \ln 4 \rightsquigarrow 4 \\ \infty \rightsquigarrow \infty \end{array} \right| = \int_4^{\infty} \frac{4}{t^2-4} = 4\int_4^{\infty} \frac{1}{t^2-4} = -4\int_4^{\infty} \frac{1}{1-\frac{t^2}{4}} = \left| \begin{array}{l} s = \frac{t}{2} \\ ds = \frac{1}{2}dt \\ 4 \rightsquigarrow 2 \\ \infty \rightsquigarrow \infty \end{array} \right| =$$

$$= -2\int_2^{\infty} \frac{1}{1-s^2}\,ds = \lim_{y\to\infty}\left(\left[-2 \operatorname{arctanh} s \right]_2^{y} \right) = \lim_{y\to\infty} \left(-2 \operatorname{arctanh} y + 2 \operatorname{arctanh} 2 \right) =$$

$$= \ln 3$$

7215-d)

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{3\cos^2 x + 1}\,dx = \left| \begin{array}{l} u = \operatorname{tg} x \\ du = \frac{1}{\cos^2 x}dx \\ -\frac{\pi}{2} \rightsquigarrow -\infty \\ \frac{\pi}{2} \rightsquigarrow \infty \end{array} \right| = \int_{-\infty}^{\infty} \frac{1}{u^2+4}\,du = 2\int_0^{\infty} \frac{1}{u^2+4}\,du = \frac{1}{2}\int_0^{\infty} \frac{1}{\frac{u^2}{4}+1}\,du = \left| \begin{array}{l} s = \frac{u}{2} \\ ds = \frac{1}{2}du \\ 0 \rightsquigarrow 0 \\ \infty \rightsquigarrow \infty \end{array} \right| =$$

$$= \int_0^{\infty} \frac{1}{s^2+1}\,ds = \lim_{a\to\infty}\left(\left[\operatorname{arctg} s \right]_0^{a} \right) = \lim_{a\to\infty} \left(\operatorname{arctg} a - \operatorname{arctg} 0 \right) = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

13/4 - b)

$$\lim_{n\to\infty}\left(\frac{n^2+1}{n^2}\right)^{n^2+1} = \lim_{n\to\infty}\left(\left(1+\frac{1}{n^2}\right)^{n^2}\cdot\left(1+\frac{1}{n^2}\right)^{1}\right) = \lim_{n\to\infty}\left(\exp\left(n^2\log\left(1+\frac{1}{n^2}\right)\right)\right) =$$

$(1+0)^1 \Rightarrow 1$

$$= \exp\left(\lim \frac{\log\left(1+\frac{1}{n^2}\right)}{\frac{1}{n^2}}\right) \overset{l'H}{=} \exp\left(\lim_{n\to\infty}\frac{1}{1+\frac{1}{n^2}}\right) = \exp\left(\frac{1}{1}\right) = e$$